

Shaun Sekura

ssekura08@gmail.com — linkedin.com/in/shaun-sekura — shaunsekura.com

EDUCATION

The Ohio State University

Columbus, OH

B.S. Mechanical Engineering, Robotics and Autonomous Systems Minor

Expected December 2027

Recent Coursework: Machine Design, Machine Elements, Kinematics, System/Vibrational Dynamics, Circuits, Fluid Mechanics, Control Systems, Sensors & Measurements, Thermodynamics, Mechanics of Materials

TECHNICAL SKILLS

- **Software:** SolidWorks, CATIA, Abaqus, Ansys, Hypermesh, Siemens Test Lab, Python, MATLAB, Excel
- **Analysis & Testing:** FEA, Modal Testing, Design of Experiments, Physical-to-Virtual Correlation, GD&T
- **Fabrication:** CNC Machining, Vertical Mill, Lathe, 3D Printing, Soldering/Wiring (Arduino), Drafting

EXPERIENCE

Mechanical Engineering Intern

Akron, OH

Hankook Tire & Technology, Virtual & Validation

Summer 2026

- Validated a revised cord material model in Abaqus finite element analysis against physically scanned tire profiles, cutting virtual-to-physical deviation 50.4% at tread grooves and to 0.86 mm at the crown.
- Developed a Python GUI to process 4,096-row tire scanner profiles with outlier filtering, windowing, and interpolated lookup, replacing a manual Excel workflow with a 71-second automated run.
- Screened 600+ cord test records to 10 qualifying datasets and differentiated least-squares curve fits ($R^2 \approx 0.99$) into strain-dependent moduli, exposing $\sim 15\%$ error in the legacy linear assumption.
- Took multiple OE wheels from CAD through Hypermesh meshing to Abaqus frequency extraction, characterizing mode shapes under bolt-hole constraints.

Undergraduate Teaching Assistant

The Ohio State University

Intro To Mechanical Design (MECHENG 2900)

Spring 2026 – Present

- Mentored 244 students per semester in circuit construction and embedded systems programming.
- Guided weekly lab sections in breadboard wiring, sensor integration, and Arduino programming.

System Design & Integration Engineer

Center For Automotive Research

EcoCAR Competition Team

Fall 2025 – Present

- Designed and simulated 10+ vehicle components in SOLIDWORKS to improve strength-to-weight ratios.
- Integrated structural designs into the final build across electronics, efficiency, and automation sub-teams.

Formula SAE

Center For Automotive Research

Student Engineer

Fall 2024 – Spring 2025

- Supported design, fabrication, and assembly of a Formula SAE race car through testing and iterative improvement.

American Society of Mechanical Engineers (ASME)

The Ohio State University

Executive Board

Fall 2024 – Spring 2026

- Organized design challenges and industry speaker events for 100+ members, raising participation 320%.

PROJECTS — SHAUNSEKURA.COM

Sustainable Rocket Redesign

- Built a reduced-order descent simulation of the rocket in Python using the AeroSandbox library.
- Reverse engineered a simplified SOLIDWORKS model of the Starship upper section.

IR-Controlled Variable-Speed Fan (Arduino)

- Programmed an Arduino to decode NEC infrared commands into three PWM speed levels with live LCD feedback.
- Designed a SOLIDWORKS enclosure and a debounced emergency-stop interlock that halts and restores fan speed.

INTERESTS

- Hiking, Music, Tennis, Basketball, Strategy Games, Competitive Trivia, Sushi